Automatic Spectral Calibration of Hyperspectral Images: Method, Dataset and Benchmark

Zhuoran Du[1,2], Shaodi You[1], Cheng Cheng[2], Shikui Wei[2]
[1]University of Amsterdam, Netherlands
[2]Visual Intelligence +X International Cooperation Joint Laboratory of MOE, Beijing Jiaotong University

Abstract

Hyperspectral images (HSI) densely sample the world in both the space and frequency domains and, therefore, are more distinctive than RGB images. Usually, HSI needs to be calibrated to minimize the impact of various illumination conditions. The traditional way to calibrate HSI utilizes a physical reference, which involves manual operations, occlusions, and/or limits camera mobility. These limitations inspire this paper to automatically calibrate HSIs using a learning-based method. Towards this goal, a large-scale HSI calibration dataset, which has 765 high-quality HSI pairs covering diversified natural scenes and illuminations, is created. The dataset is further expanded to 7650 pairs by combining with 10 different physically measured illuminations. A spectral illumination transformer (SIT) together with an illumination attention module is proposed. Extensive benchmarks demonstrate the SoTA performance of the proposed SIT. The benchmarks also indicate that low-light conditions are more challenging than normal conditions. The dataset and codes are available online: `https://github.com/duranze/Automatic-spectral-calibration-of-HSI`.

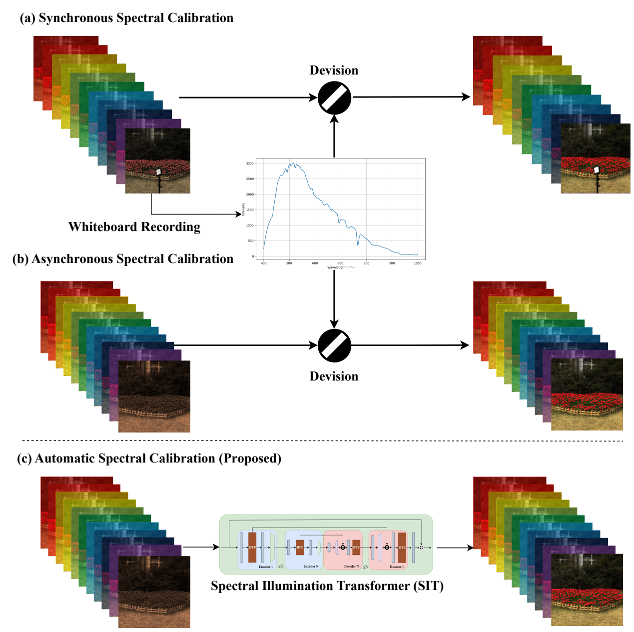


Figure 1. Comparison of HSI Calibration Methods: (a) Synchronous, (b) Asynchronous, and (c) Automatic (Proposed).

1. Introduction

Hyperspectral image (HSI) densely samples the world in both spatial and frequency domains. It is more informative than RGB images for recognizing material. Therefore it is widely used in remote sensing and laboratory analysis [21][4][6][17].

However, HSI appearance is highly dependent on global illumination. While in remote sensing and lab environments, global illuminations are highly uniform in a large spatial and temporal range or can be well controlled; global illuminations in natural scenes are highly varied spatially and temporally. Therefore, to capture HSI in natural scenes, global illumination needs to be measured simultaneously.

There are two traditional ways to measure the global illumination. The synchronous method [3] (Fig. 1.a) utilizes a reference such as a barium sulfate panel (an ideal white panel) [16] in the scene. The disadvantage is the panel partly occludes the scene. The asynchronous method [2, 8] (Fig. 1.b) takes two HSIs of the same scenes quickly with and without the reference. The disadvantage is that this method requires double shots, manual operations, and an unmoved camera.

These disadvantages inspire this paper: *can we learn to automatically calibrate the HSI without a physical reference?* If the accuracy is close to the physical methods, then the learning-based method has the advantages of less manual operations, no occlusion, and more mobility.

To enable this research, we have created the first HSI auto calibration dataset *BJTU-UVA Dataset*. The dataset is captured using a 204 bands 400-1000nm hyperspectral camera with a 512 × 512 spatial resolution, and 12 bits quantization. The dataset has 765 calibrated and uncalibrated image pairs which are obtained using the asynchronous method. The dataset covers diversified scenes and illuminations. To further study the illumination calibration, an expansion dataset *BJTU-UVA-E* is created, which composites calibrated HSIs with 10 different physically measured illuminations. In such a way, the BJTU-UVA-E has 7650 uncalibrated HSIs. It is worth noticing that BJTU-UVA is not only the first HSI calibration dataset, it is also one of the largest public HSI datasets.

After that, we propose the Spectral Illumination Transformer (SIT), which is the first learning-based HSI calibration method to the best of our knowledge. SIT is based on a U-shaped transformer. To allow SIT to learn illumination properly, we propose the *illumination attention* module which is inspired by the classic Gray-World[5] algorithm. We consider SIT the baseline of learning-based methods which may inspire future research in this new topic.

Detailed benchmarks are provided using the BJTU-UVA dataset. Given that there are no other learning-based methods for this novel task, methods for HSI denoising and reconstruction are compared. And the classic method of Gray-World is compared. It is shown that SIT is the state-of-the-art. It is interesting to find that Gray-World performs comparable to other learning-based methods on real illuminations. It is also noticed that low-light scenes/areas are likely to have larger errors; which can be a future research topic.

The novelty of the paper can be summarized as:

- We propose a novel research topic, automatic HSI calibration, which has no occlusion, less manual operations, and more mobility than traditional methods.
- We propose the large-scale HSI dataset BJTU-UVA, which is the first dataset for HSI auto-calibration.
- We propose the spectral illumination transformer (SIT) with the illumination attention module, which is the very first transformer-based method on this topic.

2. Related Work

2.1. Hyperspectral Image Calibration

In the literature, several spectral calibrations using physical references are developed for remote sensing and indoor optics.

In remote sensing, calibration is done using a physical reference (Fig. 1.a). Pan et al. in 2001 [25] used a spectroradiometer calibrated against a barium sulfate panel (the physical reference), assuming constant solar irradiance; Ferreira et al. in 2004 [11] used linear interpolation of pre- and post-flight reference panel readings; and Miura et al. in 2008 [22] used continuous ground-based panel readings during flight to adjust data. The limitation of applying this method to natural scenes is that there always exists occlusion from the white reference.

In laboratory settings, where illumination can be precisely controlled, calibration can be performed asynchronously (Fig. 1.b). Under these conditions, illumination only needs to be measured once and can then be applied in subsequent measurements. Recently, Moghadam et al. [23] proposed a data-driven method to enable indoor hyperspectral imaging using affordable and commonly available LED and fluorescent lighting sources. However, in the natural outdoor scenes, controlling illumination is infeasible. Therefore, to apply asynchronous methods, after measuring the illumination, the HSI has to be taken as soon as possible. This requires manual operations and a stable camera.

2.2. RGB White Balancing

RGB white balancing is a similar topic but with major differences. First, each of the RGB channels is a wide-band channel whereas in HSI the channel is a narrow band. Second, RGB bands usually overlap with each other whereas HSI bands do not or have limited overlaps. Third, many RGB-based methods aim for optimizing human perception while HSI calibration aims for physical accuracy.

Recent advances in white balancing for sRGB images have primarily leveraged deep learning techniques to enhance color correction. The two-stage framework by Farghaly et al. [10] combined global color mapping with local adjustments, achieving perceptually improved results. Stability across varying color temperatures has been addressed by Li et al. [18] with their SWBNet, which learned color temperature-insensitive features and applied a novel contrastive loss for consistent outcomes. Addressing the complexity of mixed-illuminant scenes, Afifi et al. [1] blended images with predefined white-balance settings using learned weighting maps, while Kinli et al. [15] modeled lighting variations as style factors, enhancing correction capabilities without explicit illuminant detection. Lastly, efficiency in processing was significantly enhanced by [14] through deterministic illumination mapping, facilitating real-time applications.

2.3. HSI Restoration and Reconstruction

Learning-based research on HSI is mainly focusing on denoising and band reconstruction. This research explores many new deep-learning architectures for HSI because it is not trivial to extend RGB-based architectures. Notice that this research does not need the HSI to be calibrated and cannot directly calibrate the HSI.

Puria et al. proposed the DivIll [23] method, which used CNN layers and residual blocks to address reflectance abnormalities caused by illumination variations in indoor scenes. Li et al. [20] introduced a local-area-based approach that linearly represented each pixel by its neighbors without requiring parameterized weight calculations, emphasizing spectrum preservation. Zhang et al. [34] provided a comprehensive review of HSI denoising techniques ranging from model-driven to hybrid model-data-driven methods. Lastly, Pan et al. [24] integrated multiscale contextual information with a coattention mechanism in the Multiscale Adaptive Fusion Network (MAFNet), showcasing improved denoising capabilities. SERT [19] introduced a rectangle self-attention mechanism to capture non-local spatial similarity and a spectral enhancement module to extract global low-rank properties of hyperspectral images for effective noise suppression. Hu *et al.* [13] proposed the HCANet by extending the U-shaped transformer to hyperspectral imaging (HSI) and leveraged a spectral transformer to capture inter-channel relationships effectively.

3. HSI Calibration

In this paper, we denote an HSI as: $I(x, y, \lambda) \in \mathbb{R}^{H\times W\times C}$, where I is the intensity, x, y are the spatial coordinates and λ the wavelength; H, W, C are the height, width, and channels correspondingly. In this paper we focus on global illumination and assume it is uniform, therefore, for simplicity, in most of this paper, we write only the wavelength $I(\lambda)$.

HSI Illumination Calibration Physically, an HSI I captures the production of scene reflectance R and illumination L wavelength-wisely:

$$I(\lambda) = R(\lambda)L(\lambda). \tag{1}$$

Unlike RGB, scene reflectance $R(\lambda)$ is highly distinctive if measured in an HSI way. And therefore it is widely used in remote sensing [21][4][6][17] and laboratory analysis[27][9]. Notice the reflectance is not albedo but still varies from shading and inter-flection and non-uniform illumination.

The goal of HSI calibration is to linearly adjust the intensity as if the global illumination is ideal white in the sampled wavelength range:

$$L(\lambda) \equiv l, \tag{2}$$

where l is a constant which usually set to be $l = 1$, then $R(\lambda) \equiv I(\lambda)$.

If the illumination $I(\lambda)$ is not ideal white, then we can calibrate the image as if the illumination is ideal white by:

$$R(\lambda) = \frac{I(\lambda)}{L(\lambda)}. \tag{3}$$

Synchronous method In a synchronized method (Fig. 1.a), $L(\lambda)$ is measured directly using a reference such as a barium sulfate panel [16]. The disadvantage is the reference has to be in the image which causes occultation.

Asynchronous method In the asynchronous method, we first measure $L_{p,t_1}(\lambda)$ on the reference at position p and time t_1. Then, take an HSI $I_{p,t_2}(\lambda)$ without the reference at the same position but at a later time. Assuming the difference between $L_{p,t_1}(\lambda)$ and $L_{p,t_2}(\lambda)$ is negligible, then: $R_2(\lambda) = \frac{I_{p,t_2}(\lambda)}{L_{p,t_1}(\lambda)}$. The disadvantages are more manual operations, and in an outdoor natural scene, the illumination can be very dynamic and therefore needs to be remeasured frequently; it also makes moving the camera infeasible.

Learning-based method (proposed) Considering the above disadvantages, we propose a novel research question: can the scene radiance be inferred directly through a learning method f without a physical reference and have satisfying accuracy as a physical measurement?

$$\tilde{R}(\lambda) = f(I(\lambda)). \tag{4}$$

If possible, it has at least three advantages: first, there is no occlusion, second, the camera can move freely, third, the imaging only needs to be done once.

4. BJTU-UVA Dataset

Our work introduces the first-ever dataset for automatic spectral calibration. The *hyperspectral automatic spectral calibration image* dataset (BJTU-UVA) comprises 765 pairs of real HSI pairs; each pair includes one original HSI and its asynchronous calibration. To further diversify the illuminations, we combine the scene radiance with ten different real illuminations to generate more uncalibrated images. The expanded dataset (BJTU-UVA-E) comprises 7650 pairs of uncalibrated and calibrated HSIs.

4.1. BJTU-UVA Dataset

For the HSI camera, we use Specim IQ [28], which offers a spectral resolution of 3nm across a wavelength range from 400nm to 1000nm. For reference, we use the IQ White Reference Board [28], which can reflect all bands of the spectrum from 400nm to 1000nm evenly.

For caption, we use the asynchronous method as described in Sec.3. Specifically, we capture the scene $I_{p,t}(\lambda)$ at position p and time t without the reference. And right after that, we capture the illumination $L_{p,t+\delta}(\lambda)$ by putting the reference in the scene. We kept the time difference small and the camera position unchanged so as to minimize the change in the illumination. The images are stored in the linear format without gamma correction in 12 bits.

In practice, the scene and illumination are not noise-free. The major source of the noise is the dark current $d(\lambda)$. Dark current refers to the electronic noise recorded by the camera's sensor even when no light is present. $d(\lambda)$ is also recorded.

The ground truth is thus generated with the consideration of $d(\lambda)$:

$$R_{gt}(\lambda) = \frac{I_{p,t}(\lambda)}{L_{p,t+\delta}(\lambda)} = \frac{I^d_{p,t}(\lambda) - d_{p,t}(\lambda)}{L^d_{p,t+\delta}(\lambda) - d_{p,t+\delta}(\lambda)}. \quad (5)$$

The dataset involves diversified scenes and illuminations as shown in Fig. 2. These scenes feature urban and natural environments under different weather conditions and times of day, including roads, buildings, statues, flowers, and other colored materials. The variety in scene composition ensures that the dataset can adequately test the efficacy of spectral calibration methods.

4.2. BJTU-UVA-E Dataset

To enable further study of the illumination. We expand the dataset by combining the ground truth reflectance $R_{gt}(\lambda)$ with 10 different physically measured illuminations $L_{S_i}(\lambda)$.

$$I_{S_i}(\lambda) = R_{gt}(\lambda) L_{S_i}(\lambda), i \in 1, 2, \cdots 10, \quad (6)$$

We record these ten different illuminations with the whiteboard as shown in Fig. 3. To minimize the dark current, we also deduct the noise $d_{S_i}(\lambda)$ from actual illuminance recording $L^d_{S_i}(\lambda)$ by

$$L_{S_i}(\lambda) = L^d_{S_i}(\lambda) - d_{S_i}(\lambda), i \in 1, 2, \cdots 10. \quad (7)$$

As shown in Fig. 3, the 10 illuminations include five natural illuminations: sunny, cloudy, rainy, evening, and shadow; which are measured using the whiteboard in the scenes. The rest five are colored filtered, which are measured by placing the color filters (red, blue, yellow, purple, and green) in front of the whiteboard.

4.3. 31 Channels Subset

Since much of the existing HSI research is conducted using 31 channels spanning 400–700 nm, we resample our HSIs to match this standard, covering 31 channels from 400–700 nm, for both the BJTU-UVA and BJTU-UVA-E datasets. This approach ensures compatibility with legacy methods.

4.4. Specifications

The dataset specifications are summarized in Fig. 2.d. Notably, this is not only the first HSI illumination calibration dataset but also one of the largest HSI datasets focused on natural scenes (further details are provided in the supplementary materials).

4.5. Evaluation Metrics

We introduce four commonly used metrics: PSNR[30], RMSE[12], ERGAS[29], and SAM[33].

The Peak Signal-to-Noise Ratio (PSNR) for HSI is defined as:

$$\text{PSNR} = \frac{1}{B} \sum_{k=1}^{B} 10 \log_{10} \left(\frac{\max \left(R^k_{gt}\right)^2}{\frac{1}{HW} \left\| R^k_{gt} - \hat{R}^k \right\|_2^2} \right), \quad (8)$$

in which B denotes the number of bands and $\|\cdot\|_2$ refers to the two-norm.

The Root Mean Squared Error (RMSE) is calculated by:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (\hat{R_{gt}}^i - R^i)^2}, \quad (9)$$

in which N denotes the number of pixels in each image.

The Error Relative Global Dimensionless Synthesis (ERGAS) metric is specifically designed for evaluating the quality of high-resolution synthesized images and can be calculated by:

$$\text{ERGAS} = \sqrt{\frac{1}{B} \sum_{k=1}^{B} \frac{\left\| R^k_{gt} - \hat{R}^k \right\|_2^2}{\mu^2 \left(R^k_{gt}\right)}}, \quad (10)$$

in which μ denotes the mean operation, and B number of bands.

The Spectral Angle Mapper (SAM) is commonly employed to assess the extent of spectral information preservation at each pixel, which can be calculated by:

$$\text{SAM} = \frac{1}{N} \sum_{i=1}^{N} \arccos \left(\frac{\langle \hat{R}^i, R^i_{gt} \rangle}{\|\hat{R}^i\|_2 \|R^i_{gt}\|_2} \right), \quad (11)$$

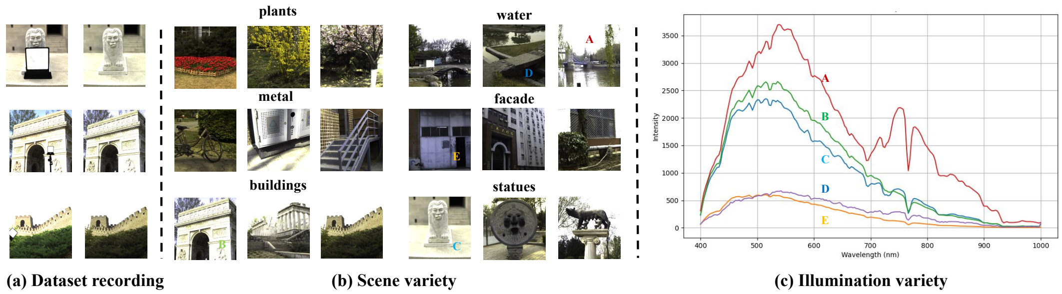
where N is the number of pixels.

5. Spectral Illumination Transformer

With the dataset ready, we propose the very first learning-based method which focuses on automatic illumination calibration for HSI: the *Spectral illumination transformer (SIT)*. We consider SIT a baseline and will inspire future research.

5.1. Framework Overview

As shown in Fig.4, the overall framework of our SIT is a U-shaped transformer proposed by [13] with residual connections. The framework is inspired by the commonly used U-Net for RGB images [26], and later extended to the transformer by Wang *et al*. [31]. Hu *et al*. [13] further extends it to HSI. The U-shaped encoder-decoder structure enables exploring more contextual information by increasing reception fields. And they use the spectral transformer to explore the relationship between channels.



Bands	204
Images	765
Spectral Range	400-1000nm
Spatial Resolution	512 x 512
Bit Depth	12
Year	2024

(d) Specification

Figure 2. Overview of the BJTU-UVA Dataset: (a) Dataset Recording Setup, (b) Variety of Scenes, (c) Illumination Variety, and (d) HSI Camera Specifications and Dataset Specification

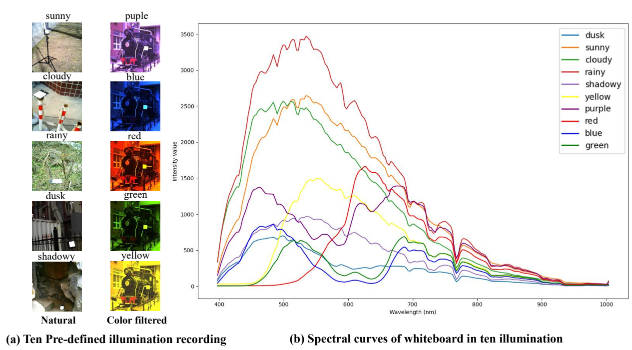


Figure 3. Measurement of Ten Different Illuminations Using a Whiteboard: Five Natural Conditions (Sunny, Cloudy, Rainy, Evening, and Shadow) and Five Color-Filtered Conditions (Red, Blue, Yellow, Purple, and Green)

However, as will be shown later by the experiments, this structure was not specially designed for HSI calibration and cannot properly capture the global illumination. Therefore, we develop a novel spectral illumination transformer unit (SIT-U).

5.2. Spectral Illumination Transformer Unit

The SIT-U is inspired by the classic RGB white balancing algorithm Gray-World [5]. The Gray-World method assumes the global mean values of all pixels in each channel are the same (*e.g.* 0.5).

As shown in Fig.4, The SIT-U is an M-layer transformer. On the m^{th} layer of SIT-U, the input feature x_0^m of the m^{th} layer of SIT-U goes through the layer normalization LN and a convolutional layer $f_{c_0}^m$ by

$$x_1^m = f_{c_0}^m(LN(x_0^m)), m = 1 \cdots M. \tag{12}$$

Then x_1^m is injected into the two branches: spectral attention (SA) calculation and illumination attention (IA) calculation. For SA calculation, we maintain the same structure as HCANet[13]. We use SA to calculate the attention map A_S^m and the attention "value" x_v^m in the m^{th} layer of SIT. Instead of immediately multiplying A_S^m with x_v^m directly as HCANet, we design the illumination attention module to import the factors of illumination into the attention process, which will be introduced in detail in the following paragraph.

Illumination Attention Inspired by the Gray-World algorithm, we propose the Illumination Attention (IA). Specifically, in the second branch "IA", we use the illumination extractor to gradually capture global illumination features x_I^m.

First, it goes through the CNN layer $f_{c_1}^m$ keeping the spatial dimension by

$$x_2^m = f_{c_1}^m(x_1^m). \tag{13}$$

Then, we gradually extract local illumination features through down-scaling convolution and average pooling AP, by

$$x_3^m = AP(f_{c_2}^m(x_2^m)), \tag{14}$$

$$x_4^m = AP(f_{c_3}^m(x_3^m)), \tag{15}$$

where the convolutional layer $f_{c_2}^m$ and $f_{c_3}^m$ have a stride of t and the pooling size of AP is p. Finally, we calculate the global mean value in each channel by

$$x_I^m = f_{avg}(x_4^m), \tag{16}$$

where f_{avg} is used to scale down the feature map x_4^m into a tensor x_I^m. This averaging operation can implicitly explore the illumination of the entire image as the Gray-World[5] method.

Then, we linearly project the illumination feature into attentive hidden space using f_l^m by

$$q_I^m = k_I^m = f_l^m(x_I^m), \tag{17}$$

where q_I^m and k_I^m are the query and key of attention model. The attention matrix A_I^m in the IA branch of m^{th} layer can be calculated by

$$A_I^m = (q_I^m)^T k_I^m. \tag{18}$$

Through the IA branch, we explore the features of global illumination and calculate the other attention matrix A_I^m.

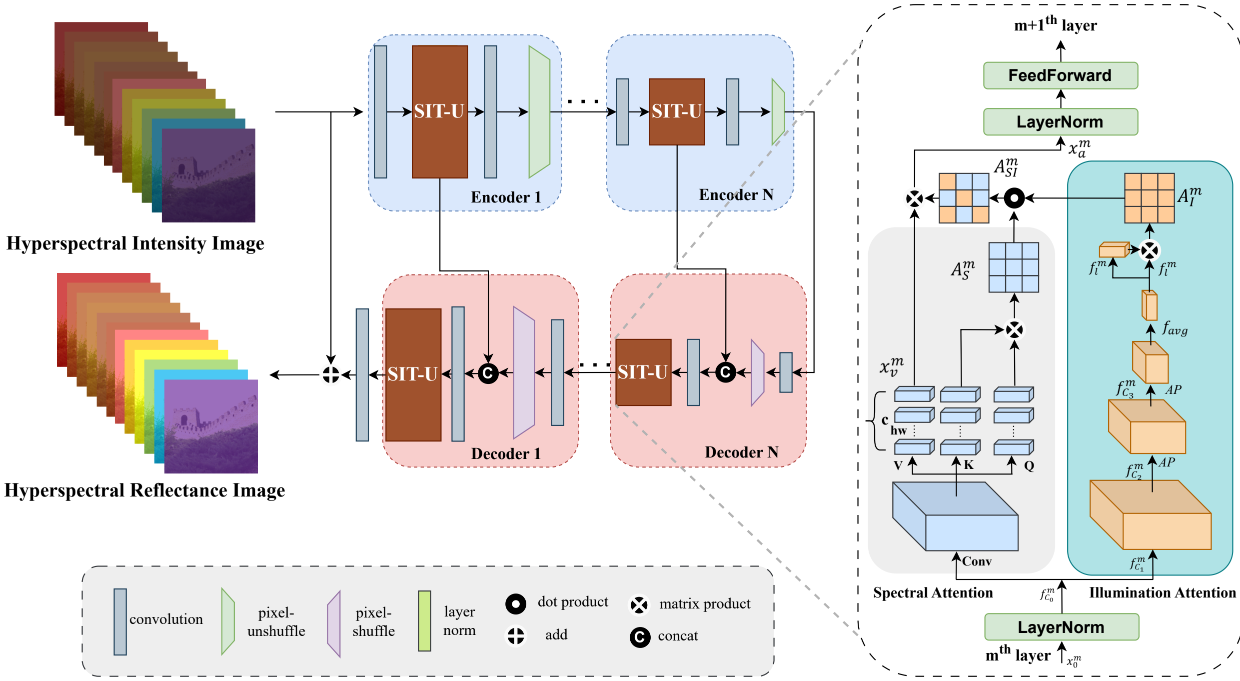


Figure 4. Structure of Spectral Illumination Transformer (SIT) framework and its unit (SIT-U) with Spectral and Illumination Attention branches for improved HSI calibration. The Illumination Attention branch mimics the Gray-World method to capture illumination features.

Spectral Illumination Attention Then we combine the two attention matrices A_S^m and A_I^m by

$$A_{SI}^m = \text{softmax}(A_S^m \cdot A_I^m), \tag{19}$$

where the A_{SI}^m is the final attention matrix in the spectral illumination attention. Then, we calculate the output of spectral illumination attention x_a^m by

$$x_a^m = A_{SI}^m x_v^m, \tag{20}$$

where the x_v^m is calculated in the spectral attention branch.

Then, as shown in Fig.4, the output x_v^m will go through layer normalization and feed-forward layers to calculate the final output of the m^{th} layer of SIT-U.

6. Experiments

6.1. Experiment Settings

The BJTU-UVA dataset is split into three sets: 535 for training, 114 for validation, and 116 for testing. Similarly, for the expansion BJTU-UVA-E: 5350 for training, 1140 for validation, and 1160 for testing.

Then we propose two tracks of benchmarks. The first is the full spectrum test (400-1000nm, 204 channels); and the second is the 31-channel test, which is to align with existing HSI-enhancement dataset and methods [32][7][3][8].

Since there are no existing learning-based methods for direct comparison, we select a classic baseline, Gray-World [5]; a recent indoor spectral restoration method, DivIll [23]; and two state-of-the-art HSI denoising methods, SERT [19] and HCANet [13].

All models in our experiments are trained on Nvidia 3090 GPUs. We randomly crop the image with a size of 256 and train with a batch size of 4. We train our model and other models for 500 epochs on the BJTU-UVA dataset and 50 epochs on the BJTU-UVA-E dataset. We use L1 loss for optimizing pixel-level differences and Adam optimizer with an initial learning rate of 10^{-4}.

6.2. Benchmarks

Comparison on Full-Spectrum HSI Table 1 presents the results. The first part shows the results on the original dataset. We observe that the Gray-World method shows relatively good performance, particularly with stable SAM scores. Among deep learning-based approaches, DivIll, SERT, and HCANet improve on metrics like PSNR, RMSE, and ERGAS; however, they struggle to significantly enhance SAM scores. In contrast, our proposed SIT method achieves the highest SAM scores across both tracks, with substantial improvements in other evaluation metrics as well.

The lower part shows the results on the expanded dataset. The Gray-World method is not comparable to the other deep learning methods on all three tracks. Although deep learning methods improve their performance on all three tracks, our proposed SIT still gets the best performance on

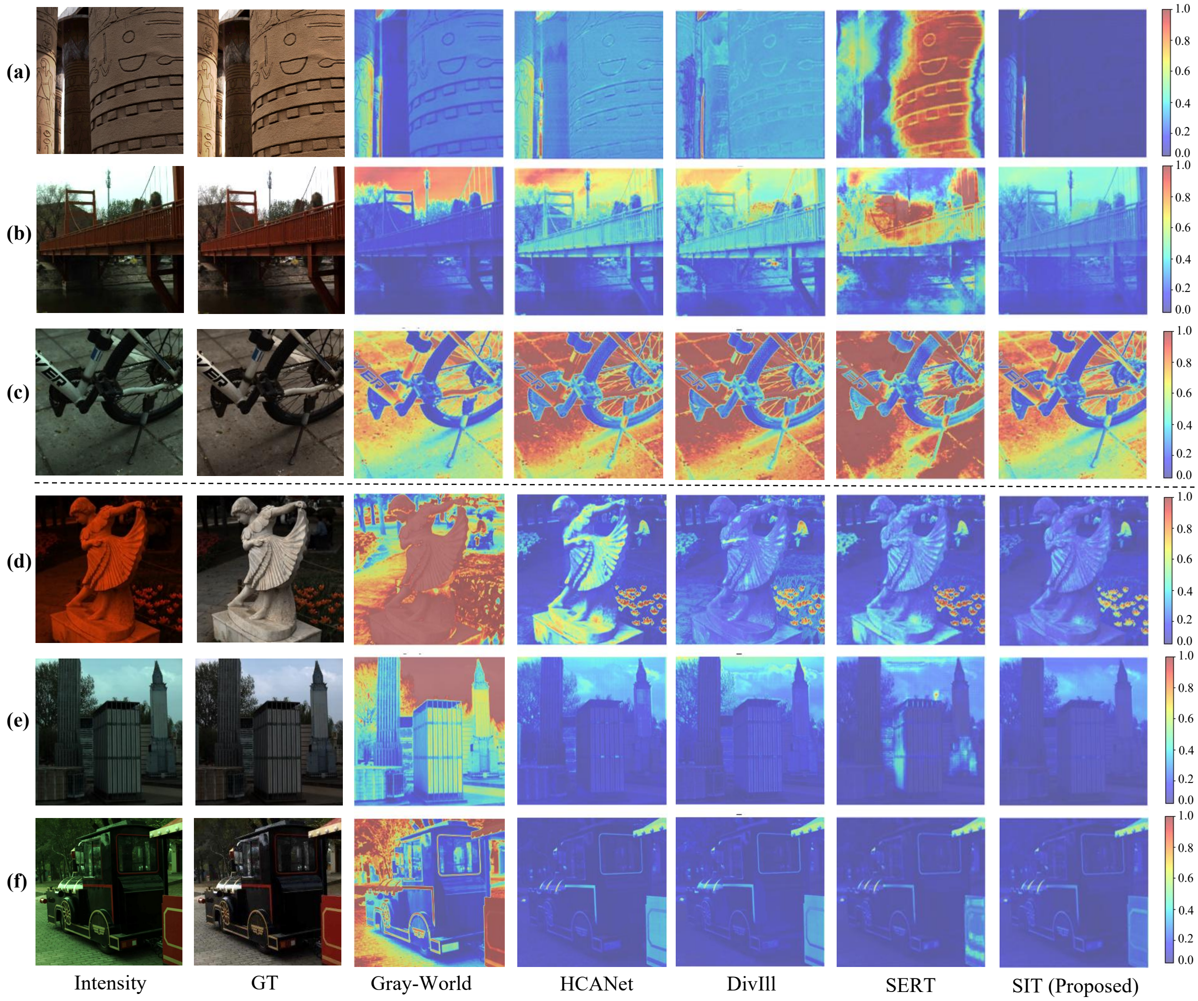


Figure 5. Visual Comparison of Absolute Error Using Heat Maps: The first two columns show the RGB rendering of the input and ground truth (GT) for visualization. Rows 1-3 display samples from BJTU-UVA, and Rows 4-6 show samples from BJTU-UVA-E.

		Val				Test			
		P↑	S↓	R↓	E↓	P↑	S↓	R↓	E↓
BJTU-UVA	Gray-World[5]	22.6	2.9	9.1	11.3	23.3	3.1	8.4	11.6
	DivIll[23]	23.5	5.5	8.3	10.2	24.1	6.1	7.6	10.0
	SERT[19]	22.3	10.4	8.3	13.2	22.8	11.1	7.9	13.5
	HCANet[13]	25.1	4.4	6.8	9.1	25.7	3.6	6.4	9.7
	SIT(proposed)	**26.1**	**2.9**	**6.2**	**8.7**	**26.3**	**3.1**	**5.8**	**9.6**
BJTU-UVA-E	Gray-World[5]	22.5	2.6	9.3	11.5	23.2	2.8	8.6	11.7
	DivIll[23]	35.2	3.1	2.2	2.7	35.2	3.4	2.2	3.0
	SERT[19]	36.4	2.7	1.8	2.6	36.4	2.9	1.7	2.7
	HCANet[13]	37.9	2.1	1.4	2.0	37.3	2.5	1.5	2.3
	SIT(proposed)	**39.3**	**1.8**	**1.2**	**1.7**	**39.1**	**2.0**	**1.3**	**1.9**

Table 1. Automatic spectral calibration evaluation on the full-spectrum HSI: PSNR(P), SAM(S), RMSE (R%), and ERGAS (E%).

all metrics.

We visualize the experimental results in Fig. 5. As can be seen, the proposed SIT is dedicated to this novel task and consistently outperforms existing methods. More results are provided in the supplementary materials.

Comparison on 31-Channel HSI Table 2 shows the performance on 31-channel HSI data. The results align with those from the full-spectrum experiments, demonstrating that the proposed SIT method achieves the best performance across all settings. Notably, all methods perform better than in the full-spectrum experiments, likely due to the reduced number of channels simplifying the tasks.

Discussions It is worth noticing that for the original datasets, which are all-natural illuminations, Gray-World is more reliable than some of the learning-based methods; it is

		Val				Test			
		P↑	S↓	R↓	E↓	P↑	S↓	R↓	E↓
BJTU-UVA	Gray-World[5]	25.2	3.3	5.8	9.6	24.2	4.1	6.5	14.2
	DivIll[23]	22.4	12.7	7.7	14.3	21.7	13.3	7.5	16.5
	SERT[19]	23.6	10.1	6.6	13.4	23.0	10.3	6.4	15.2
	HCANet[13]	26.4	3.5	5.6	9.1	26.3	3.4	5.0	11.0
	SIT(proposed)	**26.8**	**2.9**	**5.3**	**9.2**	**26.7**	**2.7**	**5.1**	**11.7**
BJTU-UVA-E	Gray-World[5]	25.3	3.1	5.8	9.5	24.2	3.6	6.5	14.1
	DivIll[23]	37.2	2.6	1.6	3.0	37.8	2.7	1.3	2.9
	SERT[19]	36.6	3.1	1.7	3.3	37.0	3.3	1.5	3.6
	HCANet[13]	37.8	2.4	1.3	2.5	38.1	2.5	1.2	2.8
	SIT(proposed)	**40.3**	**1.8**	**1.0**	**1.8**	**41.1**	**1.9**	**0.8**	**1.9**

Table 2. Automatic spectral calibration evaluation on 31-Channel HSI:PSNR(P), SAM(S), RMSE (R%), and ERGAS (E%).

	Validation				Test			
	P↑	S↓	R↓	E↓	P↑	S↓	R↓	E↓
W/O SA & IA	21.4	11.5	9.3	15.1	22.1	8.5	11.9	15.6
IA	25.5	4.1	6.3	9.2	25.8	4.3	5.9	9.6
SA	25.1	4.4	6.8	9.1	25.7	3.6	6.4	9.7
IA & SA	**26.1**	**2.9**	**6.2**	**8.7**	**26.3**	**3.1**	**5.8**	**9.6**

Table 3. Ablation Study on BJTU-UVA Full-Spectrum Data: Performance Comparison of Different Attention Configurations (Spectral Attention (SA) and Illumination Attention (IA))

reasonable because the "Gray-World" assumption is an assumption on the illumination of natural images. As for the expansion dataset, when the illuminations are not directly captured from the natural scenes, the performance is lower than other learning-based methods.

It is also noticed that learning-based methods perform significantly better on the expansion dataset than on the original dataset, whereas Gray-World does not have such improvements. This is probably because: first, the expansion dataset is much larger; second, the diversified illuminations allow the learning-based methods to better learn and model the illuminations.

6.3. Ablation Study

We study how useful the proposed illumination attention (IA) module is. Four experimental setups are evaluated: "W/O SA & IA", where only CNN layers are used; Only "SA", is a setting which falls back to HCANet [13]; "IA", where attention similarity calculation uses only IA; and "IA & SA", our complete SIT approach, where the attention maps of IA and SA are combined in each layer.

As shown in Table 3, IA can significantly improve the performance with and without the presence of SA. When both SA and IA are present, the performance is optimal. Besides, it is interesting to find that although the structure is different, the performance of SA and IA are similar when working separately.

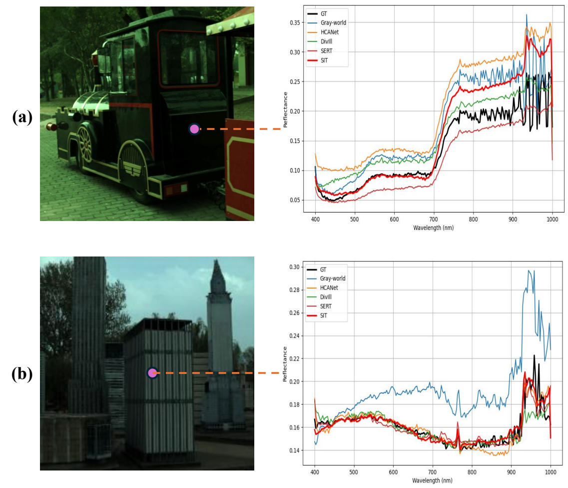


Figure 6. Challenging Cases: Comparison of Calibrated Spectra by Different Methods

6.4. Challenging Cases

As can be seen in Fig. 6, although the proposed SIT performs relatively better than other methods, it still faces challenges in low-light scenes and areas. This challenge is more significant in the infrared bands, where the illumination is generally less intense than the visible bands. We consider handling low-light scenes as a potential future work.

7. Conclusion

This paper introduces a learning-based approach for the automatic calibration of hyperspectral images (HSIs). A large-scale HSI calibration dataset, BJTU-UVA, has been developed, comprising 765 high-quality HSI pairs that encompass diverse natural scenes and illumination conditions. This dataset is further augmented to 7,650 pairs by incorporating 10 distinct, physically measured illuminations. We propose a spectral illumination transformer (SIT) with an illumination attention module, achieving state-of-the-art performance, as demonstrated by extensive benchmarks. An ablation study highlights the effectiveness of the illumination attention module. The benchmark also reveals limitations in low-light areas across both visible and infrared bands, suggesting a direction for future work.

Acknowledgement

This work was conducted during the author's visit to the University of Amsterdam, Netherlands (UvA), partly supported by the Chinese Scholarship Council (CSC). Additionally, this work was supported by the National Key R&D Program of China (2024YFC3014900).

References

[1] Mahmoud Afifi, Marcus A Brubaker, and Michael S Brown. Auto white-balance correction for mixed-illuminant scenes. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 1210–1219, 2022. 2

[2] Boaz Arad and Ohad Ben-Shahar. Sparse recovery of hyperspectral signal from natural rgb images. In *European Conference on Computer Vision*, pages 19–34. Springer, 2016. 1

[3] Boaz Arad, Radu Timofte, Rony Yahel, Nimrod Morag, Amir Bernat, Yuanhao Cai, Jing Lin, Zudi Lin, Haoqian Wang, Yulun Zhang, et al. Ntire 2022 spectral recovery challenge and data set. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 863–881, 2022. 1, 6

[4] E Ben-Dor, S Chabrillat, and José AM Demattê. Characterization of soil properties using reflectance spectroscopy. In *Fundamentals, sensor systems, spectral libraries, and data mining for vegetation*, pages 187–247. CRC press, 2018. 1, 3

[5] Gershon Buchsbaum. A spatial processor model for object colour perception. *Journal of the Franklin institute*, 310(1): 1–26, 1980. 2, 5, 6, 7, 8

[6] Simon Carter, Andy Fisher, Bridget Gibson, John Marshall, Ben Russell, and Ian Whiteside. Atomic spectrometry update: review of advances in the analysis of metals, chemicals and materials. *Journal of Analytical Atomic Spectrometry*, 32(11):2068–2117, 2017. 1, 3

[7] Ayan Chakrabarti and Todd Zickler. Statistics of real-world hyperspectral images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 193–200. IEEE, 2011. 6

[8] Zhuoran Du, Shikui Wei, Ting Liu, Shunli Zhang, Xiaotong Chen, Shiyin Zhang, and Yao Zhao. Exploring the applicability of spectral recovery in semantic segmentation of rgb images. *IEEE Transactions on Multimedia*, 2023. 1, 6

[9] Gamal M ElMasry and Shigeki Nakauchi. Image analysis operations applied to hyperspectral images for non-invasive sensing of food quality–a comprehensive review. *Biosystems engineering*, 142:53–82, 2016. 3

[10] Marwa Farghaly, Romany F Mansour, and Adel A Sewisy. Two-stage deep learning framework for srgb image white balance. *Signal, Image and Video Processing*, 17(1):277–284, 2023. 2

[11] LG Ferreira, H Yoshioka, A Huete, and EE Sano. Optical characterization of the brazilian savanna physiognomies for improved land cover monitoring of the cerrado biome: preliminary assessments from an airborne campaign over an lba core site. *Journal of Arid Environments*, 56(3):425–447, 2004. 2

[12] Rafael C Gonzalez and Richard E Woods. *Digital Image Processing*. Prentice Hall, 2nd edition, 2002. 4

[13] Shuai Hu, Feng Gao, Xiaowei Zhou, Junyu Dong, and Qian Du. Hybrid convolutional and attention network for hyperspectral image denoising. *IEEE Geoscience and Remote Sensing Letters*, 2024. 3, 4, 5, 6, 7, 8

[14] Furkan Kınlı, Doğa Yılmaz, Barış Özcan, and Furkan Kıraç. Deterministic neural illumination mapping for efficient auto-white balance correction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1139–1147, 2023. 2

[15] Furkan Kınlı, Doğa Yılmaz, Barış Özcan, and Furkan Kıraç. Modeling the lighting in scenes as style for auto white-balance correction. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 4903–4913, 2023. 2

[16] George A Leshkevich. Goniometric measurements of a spray-painted barium sulfate reference panel. *Remote Sensing of Environment*, 24(2):287–296, 1988. 1, 3

[17] Chunyu Li, Yusuke Monno, and Masatoshi Okutomi. Spectral mvir: Joint reconstruction of 3d shape and spectral reflectance. In *2021 IEEE International Conference on Computational Photography (ICCP)*, pages 1–12. IEEE, 2021. 1, 3

[18] Chunxiao Li, Xuejing Kang, Zhifeng Zhang, and Anlong Ming. Swbnet: a stable white balance network for srgb images. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 1278–1286, 2023. 2

[19] Miaoyu Li, Ji Liu, Ying Fu, Yulun Zhang, and Dejing Dou. Spectral enhanced rectangle transformer for hyperspectral image denoising. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5805–5814, 2023. 3, 6, 7, 8

[20] Shouzhi Li, Liangliang Zhu, Luyan Ji, Xin Yi, Yongchao Zhao, and Xiurui Geng. Hyperspectral image denoising based on the similar spectra approaching. *Infrared Physics & Technology*, 131:104667, 2023. 2

[21] Maria Losurdo, Michael Bergmair, Giovanni Bruno, Denis Cattelan, Christoph Cobet, Antonello De Martino, Karsten Fleischer, Zorana Dohcevic-Mitrovic, Norbert Esser, Melanie Galliet, et al. Spectroscopic ellipsometry and polarimetry for materials and systems analysis at the nanometer scale: state-of-the-art, potential, and perspectives. *Journal of Nanoparticle Research*, 11:1521–1554, 2009. 1, 3

[22] Tomoaki Miura, Alfredo R Huete, Laerte Guimaães Ferreira, Edson E Sano, and Hiroki Yoshioka. A technique for reflectance calibration of airborne hyperspectral spectrometer data using a broad, multiband radiometer. In *Hyperspectral Remote Sensing of Tropical and Sub-tropical Forests*, pages 213–232. CRC Press, 2008. 2

[23] Puria Azadi Moghadam, Neha Sharma, and Mohamed Hefeeda. Enabling hyperspectral imaging in diverse illumination conditions for indoor applications. In *Proceedings of the 12th ACM Multimedia Systems Conference*, pages 23–35, 2021. 2, 6, 7, 8

[24] Haodong Pan, Feng Gao, Junyu Dong, and Qian Du. Multiscale adaptive fusion network for hyperspectral image denoising. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16:3045–3059, 2023. 3

[25] Lorraine A Remer, Andrew E Wald, and Yoram J Kaufman. Angular and seasonal variation of spectral surface reflectance ratios: Implications for the remote sensing of aerosol over

land. *IEEE Transactions on Geoscience and Remote Sensing*, 39(2):275–283, 2001. 2

[26] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18*, pages 234–241. Springer, 2015. 4

[27] Eric Ariel L Salas and Geoffrey M Henebry. A new approach for the analysis of hyperspectral data: Theory and sensitivity analysis of the moment distance method. *Remote sensing*, 6 (1):20–41, 2013. 3

[28] Specim, Spectral Imaging Ltd. Specim - Hyperspectral Imaging Cameras and Systems. https://www.specim.com/. 3

[29] Lucien Wald. Quality of high resolution synthesized images: Is there a simple criterion? In *Proceedings of the Third Conference "Fusion of Earth Data"*, pages 99–103, 2000. 4

[30] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004. 4

[31] Zhendong Wang, Xiaodong Cun, Jianmin Bao, Wengang Zhou, Jianzhuang Liu, and Houqiang Li. Uformer: A general u-shaped transformer for image restoration. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 17683–17693, 2022. 4

[32] Fumihito Yasuma, Tomoo Mitsunaga, Daisuke Iso, and Shree K Nayar. Generalized assorted pixel camera: post-capture control of resolution, dynamic range, and spectrum. *IEEE transactions on image processing*, 19(9):2241–2253, 2010. 6

[33] Richard H Yuhas, Alexander FH Goetz, and Joe W Boardman. Discrimination among semi-arid landscape endmembers using the spectral angle mapper (sam) algorithm. In *Summaries of the Third Annual JPL Airborne Geoscience Workshop*, pages 147–149, 1992. 4

[34] Qiang Zhang, Yaming Zheng, Qiangqiang Yuan, Meiping Song, Haoyang Yu, and Yi Xiao. Hyperspectral image denoising: From model-driven, data-driven, to model-data-driven. *IEEE Transactions on Neural Networks and Learning Systems*, 2023. 2